

Evaluation of the Utility of Some PIT Tag Sites

Wenatchee

Kevin E. See

2026-05-04

Table of contents

Goal	1
Methods	1
Results	3
Tag Detections	3
Detection Probability	4
Escapement	8
Discussion	11

Goal

Our goal is to evaluate the impact of removing several PIT tag detection sites in the Wenatchee basin on steelhead escapement estimates made with the Dam Adult Branch Occupancy Model (DABOM). Specifically, we examined the effects of removing CHL (in the upper Chiwawa River), NAL (in upper Nason Creek) and PEU (in upper Peshastin Creek).

Methods

We used the exact same PIT tag detections that were used in the latest DABOM estimates. We chose several years to test across a range of escapement sizes (2011, 2014, 2017, 2022). We removed any detections from CHL, NAL or PEU, and dropped those three sites from

the model. We then re-ran DABOM and compiled estimates of detection probability and abundance at the site immediately downstream of those sites to compare with the original estimates (Table 1). We plotted the differences, and calculated summary statistics.

Table 1: Sites evaluated for removal, the closest downstream site to them and on what tributary they are located.

Removed Site	Downstream Site	Tributary
CHU	CHL	Chiwawa River
NAU	NAL	Nason Creek
PEU	PES	Peshastin Creek

Results

Tag Detections

Table 2: Number of tags detected at the lower site, the upper site, and the upper site only for each year and tributary.

Spawn Year	Stream	N Lower	N Upper	N Upper Only
2011	Chiwawa	17	10	2
2011	Nason	33	21	6
2011	Peshastin	12	0	0
2014	Chiwawa	31	23	5
2014	Nason	53	18	0
2014	Peshastin	32	0	0
2017	Chiwawa	6	4	1
2017	Nason	11	6	1
2017	Peshastin	7	3	1
2022	Chiwawa	1	2	1
2022	Nason	4	3	1
2022	Peshastin	8	0	0

Detection Probability

Table 3: Estimates of detection probability, with and without including detections from the upper site.

Spawn Year	Node	N Tags	Est w Upper	Est wo Upper	Error	Relative Err (%)
2011	CHL_D	14	0.68 (0.1)	0.75 (0.11)	0.063	9.2
2011	CHL_U	14	0.69 (0.11)	0.75 (0.11)	0.062	9.0
2011	NAL_D	28	0.71 (0.07)	0.84 (0.06)	0.124	17.4
2011	NAL_U	32	0.82 (0.06)	0.95 (0.04)	0.136	16.6
2011	PES_D	9	0.68 (0.13)	0.68 (0.13)	-0.004	-0.6
2011	PES_U	11	0.83 (0.12)	0.82 (0.11)	-0.007	-0.8
2014	CHL_D	26	0.68 (0.08)	0.78 (0.08)	0.097	14.3
2014	CHL_U	25	0.65 (0.08)	0.75 (0.08)	0.094	14.5
2014	NAL_D	39	0.72 (0.06)	0.71 (0.06)	-0.009	-1.3
2014	NAL_U	51	0.94 (0.04)	0.93 (0.04)	-0.008	-0.8
2014	PES_D	29	0.88 (0.06)	0.89 (0.06)	0.005	0.6
2014	PES_U	31	0.94 (0.05)	0.94 (0.05)	0.004	0.4
2017	CHL_D	4	0.44 (0.17)	0.28 (0.18)	-0.162	-37.1
2017	CHL_U	3	0.34 (0.16)	0.21 (0.16)	-0.131	-38.0
2017	NAL_D	10	0.78 (0.11)	0.83 (0.11)	0.054	6.9
2017	NAL_U	10	0.78 (0.11)	0.84 (0.11)	0.060	7.7
2017	PES_D	6	0.68 (0.14)	0.77 (0.14)	0.087	12.8
2017	PES_U	7	0.77 (0.13)	0.89 (0.12)	0.118	15.2
2022	CHL_D	1	0.33 (0.22)	1 (0)	0.668	201.0
2022	CHL_U	0	0 (0)	0 (0)	0.000	-
2022	NAL_D	4	0.66 (0.18)	0.71 (0.2)	0.055	8.4
2022	NAL_U	1	0.23 (0.15)	0.24 (0.16)	0.008	3.3
2022	PES_D	5	0.54 (0.16)	0.52 (0.15)	-0.018	-3.3
2022	PES_U	7	0.74 (0.14)	0.73 (0.15)	-0.010	-1.4

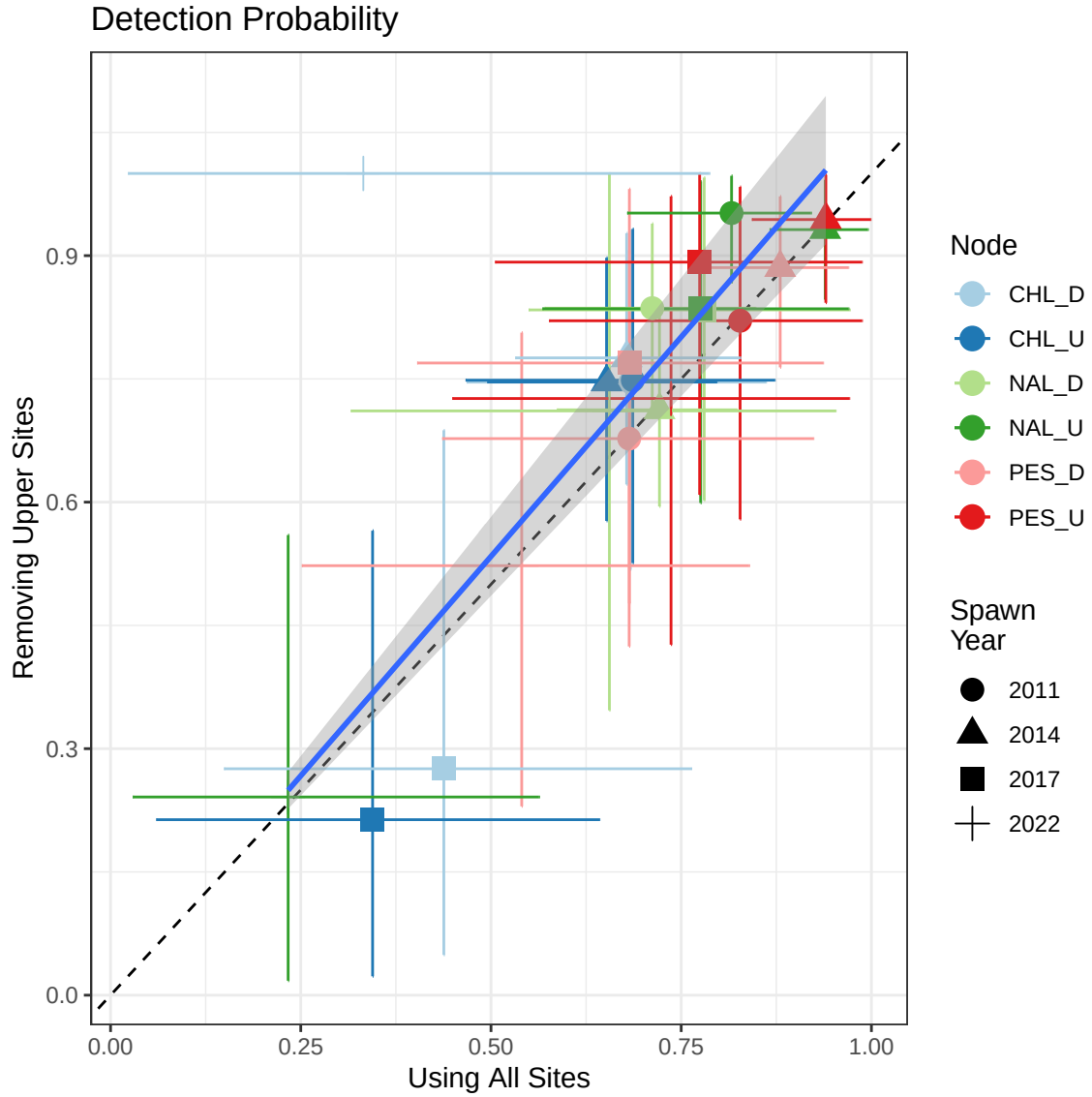


Figure 1: Estimates of detection probabilities at each node from model runs that included the upstream sites (x-axis) and those where those sites were removed (y-axis). Colors correspond to the arrays, shape to the spawn year and the whiskers represent 95% confidence intervals. There is a dashed one-to-one line, and a blue linear regression line with a fixed intercept at zero is also shown.

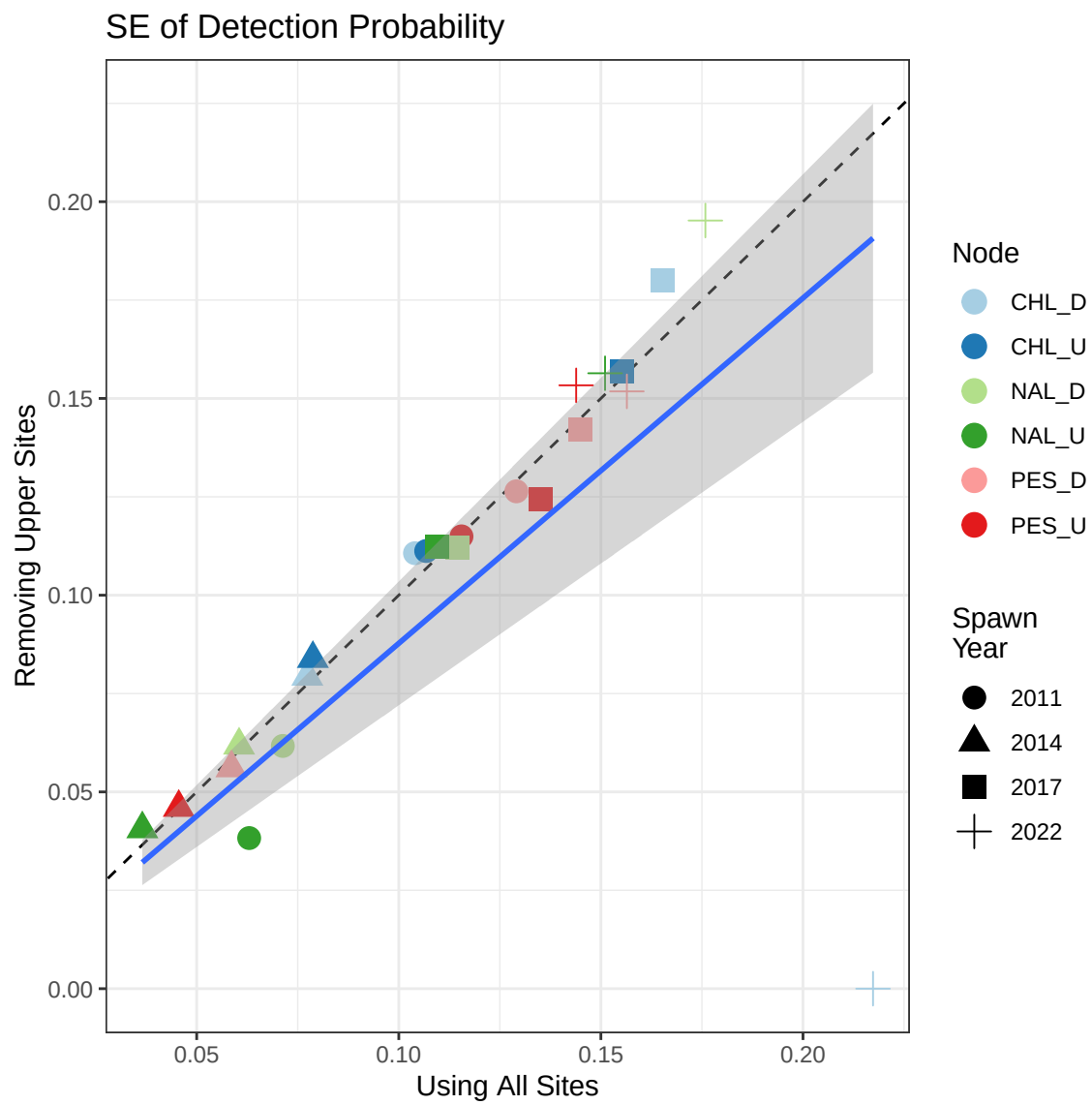


Figure 2: Plot of the standard error of the detection probability from model runs that included the upstream sites (x-axis) and those where those sites were removed (y-axis). Colors correspond to the arrays and shape to the spawn year. There is a dashed one-to-one line, and a blue linear regression line with a fixed intercept at zero is also shown.

Table 4: Summary statistics of the detection probability errors associated with model runs without the upper sites. Statistics include mean error (ME) mean relative error (MRE), mean absolute error (MAE), mean absolute percent error (MAPE) and root mean squared error (RMSE).

Node	ME	MRE (%)	MAE	MAPE (%)	RMSE
CHL_D	-0.001	-4.5	0.108	20.2	0.115
CHL_U	0.008	-4.8	0.096	20.5	0.100
NAL_D	0.056	7.9	0.061	8.5	0.073
NAL_U	0.049	6.7	0.053	7.1	0.074
PES_D	0.018	2.4	0.029	4.3	0.045
PES_U	0.026	3.4	0.035	4.5	0.059
Overall	0.056	11.1	0.086	18.3	0.159

Escapement

Table 5: Estimates of escapement, with and without including detections from the upper site.

Spawn Year	Location	Origin	Est w Upper	Est wo Upper	Error	Relative Err (%)
2011	CHL	H	52 (26)	55 (27)	2.5	4.802
2011	CHL	W	181 (47)	157 (45)	-24.1	-13.315
2011	NAL	H	207 (47)	165 (43)	-42.4	-20.465
2011	NAL	W	193 (46)	172 (41)	-21.7	-11.232
2011	PES	H	44 (25)	44 (23)	-0.5	-1.150
2011	PES	W	114 (37)	116 (37)	2.1	1.889
2014	CHL	H	103 (25)	92 (25)	-11.6	-11.235
2014	CHL	W	131 (28)	113 (27)	-17.4	-13.289
2014	NAL	H	139 (29)	139 (30)	0.5	0.353
2014	NAL	W	182 (34)	182 (33)	-0.5	-0.296
2014	PES	H	10 (9)	10 (9)	-0.2	-1.625
2014	PES	W	196 (36)	196 (34)	-0.7	-0.357
2017	CHL	H	45 (22)	74 (48)	28.7	64.037
2017	CHL	W	18 (12)	22 (25)	4.1	23.386
2017	NAL	H	37 (15)	37 (15)	0.0	0.136
2017	NAL	W	36 (15)	31 (14)	-4.7	-13.188
2017	PES	H	5 (6)	4 (6)	-0.6	-12.688
2017	PES	W	55 (19)	48 (19)	-6.7	-12.243
2022	CHL	H	6 (9)	4 (5)	-2.1	-37.143
2022	CHL	W	26 (21)	11 (10)	-14.5	-56.057
2022	NAL	H	2 (5)	2 (5)	0.0	1.235
2022	NAL	W	40 (19)	37 (19)	-2.7	-6.848
2022	PES	H	13 (11)	14 (12)	0.5	4.208
2022	PES	W	68 (24)	70 (26)	2.1	3.068

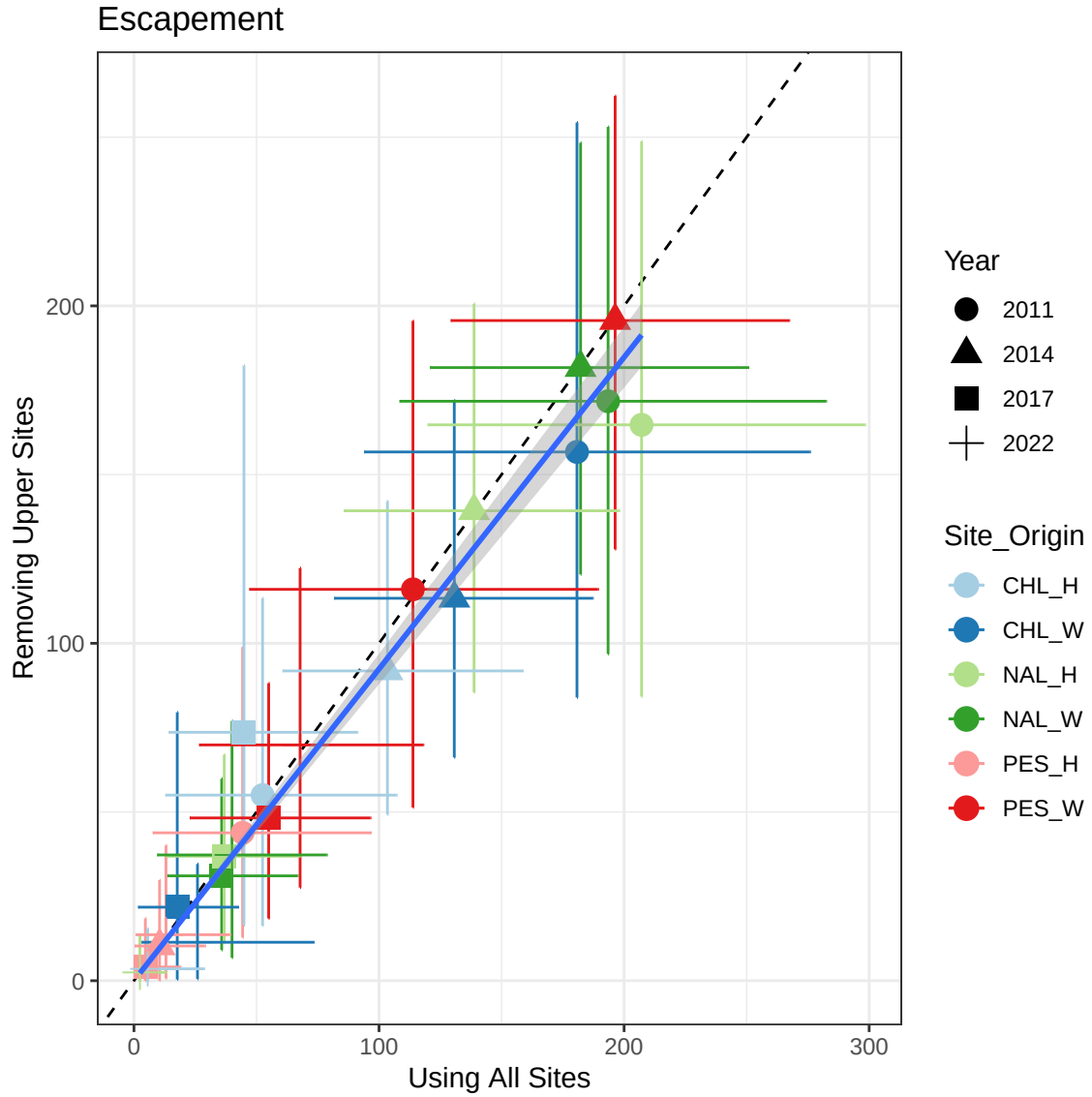


Figure 3: Estimates of escapement past each site for model runs that included the upstream sites (x-axis) and those where those sites were removed (y-axis). Colors correspond to the combination of site and origin (W for natural-origin, H for hatchery-origin), shape to the spawn year and the whiskers represent 95% confidence intervals. There is a dashed one-to-one line, and a blue linear regression line with a fixed intercept at zero is also shown.

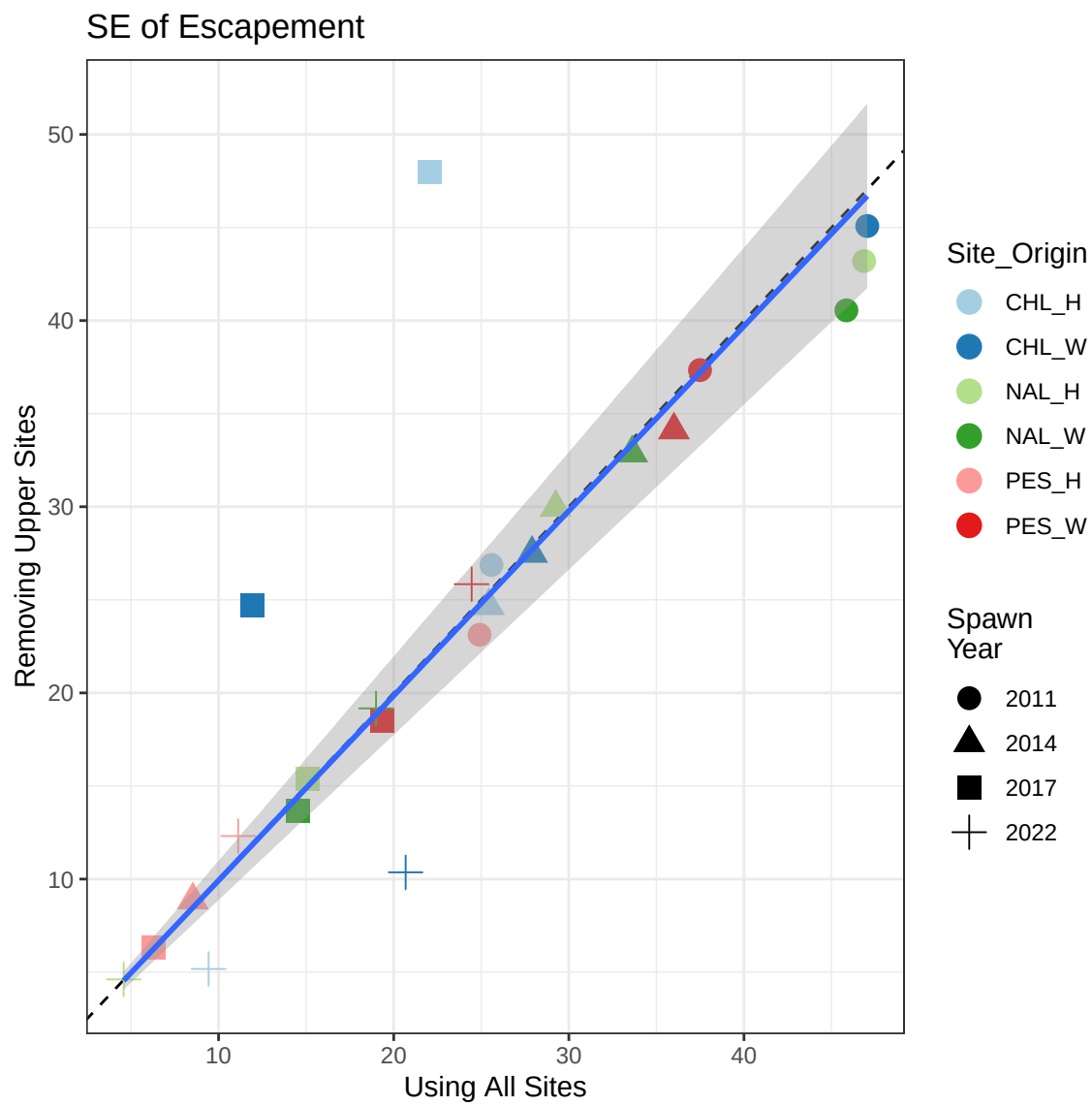


Figure 4: Plot of the standard error of the escapement estimates from model runs that included the upstream sites (x-axis) and those where those sites were removed (y-axis). Colors correspond to the arrays and shape to the spawn year. There is a dashed one-to-one line, and a blue linear regression line with a fixed intercept at zero is also shown.

Table 6: Summary statistics of the escapement errors associated with model runs without the upper sites. Statistics include mean error (ME) mean relative error (MRE), mean absolute error (MAE), mean absolute percent error (MAPE) and root mean squared error (RMSE).

Location	Origin	ME	MRE (%)	MAE	MAPE (%)	RMSE
CHL	H	6.547	19.2	14.293	26.7	17.957
CHL	W	-12.437	-1.1	15.190	16.7	17.303
NAL	H	-10.455	-4.7	10.740	5.5	21.196
NAL	W	-7.432	-7.9	7.432	7.9	11.206
PES	H	-0.180	-2.8	0.455	4.9	0.485
PES	W	-0.800	-1.9	2.915	4.4	3.699
Overall	H	-2.082	-0.8	7.478	13.3	15.193
Overall	W	-7.064	-8.2	8.457	12.9	11.784

Discussion

Overall, removing these three upper sites had little effect on detection probability and escapement estimates. In general, the updated detection probabilities were sometimes biased slightly high, leading to escapement estimates that were biased slightly low. In the Chiwawa River in 2022, there were no detections on the CHL_U node, which forced the model to fix detection probability of the CHL_D node to 100%, leading to an underestimate of escapement to the Chiwawa. When the upper site, CHU, was included, detection probability of CHL_D could be estimated at 33% (Table 3).

An upper site provides the most information to the model when it contains detections of tags that were not detected at the lower site. This happened only a handful of times in our example years (Table 2). When the upper site did not detect any tags that weren't also detected at the lower site, the estimates of detection probability and escapement at the lower site weren't affected at all (other than slight variation due to MCMC sampling runs).

If the upper sites are removed, it is important for these results that the lower site be maintained such that the two arrays are as independent as possible, to meet the assumptions of DABOM.